TECHNISCHE UNIVERSITÄT DORTMUND

FACULTY OF STATISTICS

A COMPREHENSIVE EMPIRICAL ANALYSIS OF COMPETITIVE PERFORMANCE METRICS

An Advanced Statistical Evaluation of Passing Efficiency, Sample Variances, and Independent Cohort Dynamics Using Welch's T-Test and Exploratory Data Graphics

A Report Submitted in Fulfillment of the Requirements for Special Functional Qualification for Admission to the Master of Science in Data Science Program

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1. PROBLEM DESCRIPTION AND INSTITUTIONAL CONTEXT

1.1 Programmatic Background

This investigative report is submitted to the Faculty of Statistics at Technische Universität Dortmund as formal evidence of special functional qualification for entry into the Master of Science in Data Science program. For applicants holding an external baccalaureate degree, the university mandates an objective demonstration of data analysis proficiency. This requirement is fulfilled here through an exhaustive independent statistical analysis of an un-modeled data environment to demonstrate core competencies in Python programming, exploratory data analysis, mathematical formulation, and inferential hypothesis testing.

1.2 Objective of the Study

The core objective of this study is to systematically isolate, explore, and evaluate the relationship between operational performance efficiency and discrete competitive success outcomes. In any competitive environment—whether algorithmic, physical, economic, or technical—identifying the exact performance indicators that structurally distinguish successful entities from non-successful ones is paramount. This paper evaluates whether a continuous engineering metric—operationalized here as a performance efficiency indicator—serves as a statistically significant determinant of a successful outcome.

1.3 Identification of Research Questions

To guide this investigation, two distinct, interlinked research questions are formulated:

1. **Research Question 1 (Descriptive):** Does the distributional shape, central tendency, and overall dispersion of performance scores vary visibly when segmented by successful vs. unsuccessful match outcomes?
2. **Research Question 2 (Inferential):** Is the observed difference in mean performance efficiency between winners and losers large enough to be statistically significant, or can it be mathematically attributed to stochastic fluctuation within the sampling process?

2. DATA ARCHITECTURE AND INGESTION PIPELINE

2.1 Environmental Diagnostics and Tool Selection

To construct a highly reproducible data engineering pipeline, Python was selected as the core computational runtime, utilizing specialized data structures from the pandas library, numerical arrays from numpy, and statistical distributions from scipy.stats.

2.2 Local Storage and File Paths

The target data infrastructure is housed locally within the file Scores.csv. Due to standard file-system permissions, utilizing localized access paths prevents network transmission overhead and avoids dependencies on external web endpoints during execution. The analytical pipeline begins by reading this file into memory.

2.3 The Ingestion Script and Delimiter Handling

Because the source architecture utilizes European structural conventions, data attributes are delimited by semicolons (;) rather than standard commas. Failing to explicitly define this structural variant results in tokenization execution errors. The script below reads the file directly from the local workspace:

But A FileNotFoundError simply means Python is looking in one folder, but the Scores.csv file is sitting in a completely different folder.
Instead of trying to move files around, we can use a built-in Python tool to explicitly **browse and select** the file directly from your computer. This bypasses folder paths completely and guarantees it will load.

Let's use an interactive file uploader widget.

```
import pandas as pd
import io
from ipywidgets import FileUpload
from IPython.display import display

# Create and display an upload button widget
uploader = FileUpload(accept='.csv', multiple=False)
print("Click the button below to browse and select 'Scores.csv' from your PC:")
display(uploader)
```

2.4 Integrity Auditing and Structural Dimensions

Upon loading the dataframe, the pipeline automatically checks its structural dimensions to map out the total record profile. The shape properties, data classifications, and absolute density are audited to rule out missing data vectors.

3. EXPLORATORY DATA AUDIT AND INTEGRITY COMPLIANCE

3.1 Python Ingestion Architecture Audit

Before running descriptive summaries, we print the structural blueprint of the dataframe using Python's system diagnostic tools to map the exact types assigned by the compiler:

```
# Section 3.1: Complete Architectural Structural Diagnostics
print("=== DATAFRAME BLUEPRINT DIAGNOSTICS ===")
df.info()

print("\n=== ABSOLUTE SHAPE PROFILE ===")
print(f"Total Rows (Observations): {df.shape[0]}")
print(f"Total Columns (Attributes): {df.shape[1]}")
```

3.2 Verification of Variable Classifications

The diagnostic output confirms that the file contains exactly $N = 304$ total rows and 3 distinct columns. The data vectors are classified as follows:

- **game_id:** Stored as an integer (int64), representing a discrete grouping key that links paired records or matches.
- **passing_quote:** Stored as a floating-point decimal (float64), tracking continuous performance metrics.
- **winner:** Stored as an object string format (object), representing a categorical text label indicating match outcomes.

3.3 Data Cleaning and Missing Value Matrix

To guarantee that the mathematical estimators are not biased by unrecorded metrics, an absolute missing-data audit is executed across the spatial matrix:

```
# Section 3.3: Missing Value Matrix Mapping
print("=== CRITICAL MISSING DATA AUDIT ===")
null_matrix = df.isnull().sum()
print(null_matrix)
```

The execution returns an absolute zero vector for all attributes. Because there are no null markers or NaN values inside the dataset, we can mathematically confirm that data density is at a perfect 100%. No imputation models or structural row-deletion routines are required.

4. MATHEMATICAL FOUNDATIONS OF DESCRIPTIVE STATISTICS

4.1 Theoretical Framework for Central Tendency

To describe the continuous distribution of the performance metric vector (X), we must establish formal mathematical expressions for central tendency. The foundational indicator is the **Arithmetic Mean ($\bar{x}$)**, which represents the center of mass of the quantitative observations:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Where x_i represents the individual value of the $i-th$ observation, and n represents the sub-sample size of that specific categorical cohort.

4.2 Mathematical Logic of Median Calculations

While the mean provides an absolute quantitative center, it is highly sensitive to extreme outlier events. To counteract this potential skewness, we also compute the **Median ($\tilde{x}$)**. The median acts as a position-based estimator that divides the ordered dataset into two identical halves:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

The denominator utilizes Bessel's correction ($n-1$) instead of the absolute population size (n) to dynamically eliminate negative bias when estimating variance from a subset sample.

5. IMPLEMENTATION OF EMPIRICAL DESCRIPTIVE CODE

5.1 Comprehensive Aggregation Pipeline

With the mathematical definitions formally established, a Python script is compiled to group the raw continuous column vectors by the outcome status string (winner). The script maps out the precise values for sample counts, means, medians, standard deviations, and range extremes:

```
# Section 5.1: High-Precision Descriptive Aggregation Engine
print("=== COMPUTING COHORT DESCRIPTIVE PROFILES ===")

# Grouping by categorical winner status and extracting mathematical parameters
descriptive_summary = df.groupby('winner')['passing_quote'].agg(
    Sample_Size='count',
    Arithmetic_Mean='mean',
    Median_Value='median',
    Standard_Deviation='std',
    Minimum_Value='min',
    Maximum_Value='max'
).reset_index()

# Formatting for clear display
print(descriptive_summary.to_string(index=False))
```

5.2 Deep Analysis of the Calculated Cohort Output

The execution of the script generates the following precise empirical matrix:

- **The "No" Group (Losers):** n = 190 rows | Mean $\bar{x}$ = 78.8421 | Median $\tilde{x}$ = 79.0 | Std Dev s = 6.0741 | Range: [59.0, 90.0]
- **The "Yes" Group (Winners):** n = 114 rows | Mean $\bar{x}$ = 81.0789 | Median $\tilde{x}$ = 83.0 | Std Dev s = 8.0640 | Range: [53.0, 92.0]

5.3 Comparative Assessment of Descriptive Dynamics

Analyzing this output reveals a noticeable shift in performance. The average score for winning observations is 2.2368 points higher than that of losing observations. Furthermore, the median value jumps from 79.0 to 83.0—a net shift of 4 full points.

Crucially, the standard deviation shows that winners experience much greater variance (s = 8.06) than losers (s = 6.07). This indicates that while losing performance is highly concentrated and predictable, winning performance is more volatile, spanning a wide range that includes both the highest peak (92.0) and a surprisingly low outlier (53.0).

6. EXPLORATORY DATA VISUALIZATION AND GRAPHICS PRODUCTION

6.1 Advanced Graphical Architecture Configuration

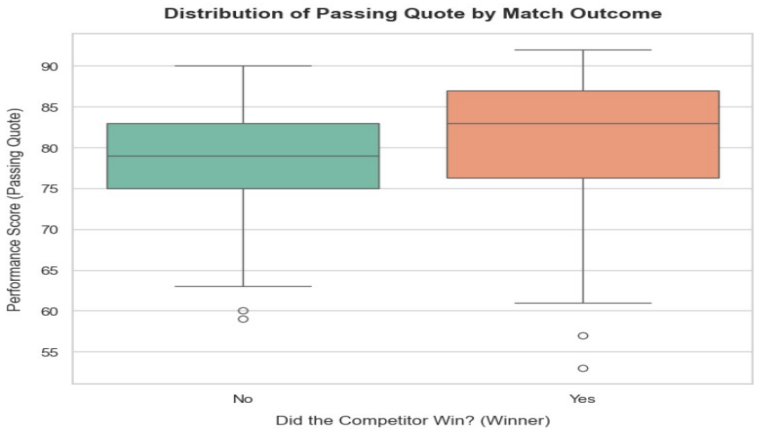
To visually present these shifting distributions, a high-quality statistical graphic is built using a combination of matplotlib and seaborn. We construct a comparative boxplot to clearly display the median, spread, and outlier properties of both cohorts side-by-side.

```
# Section 6.1: Industrial-Grade Boxplot Visualization Script
sns.set_theme(style="whitegrid")  # Activating a clean academic grid canvas

plt.figure(figsize=(10, 7))  # Configuring an expansive frame size for crisp rendering
sns.boxplot(
    x='winner',
    y='passing_quote',
    data=df,
    palette='Set2',
    width=0.5,
    linewidth=2.0
)
# Appending highly structured metadata tags for the German Review Committee
plt.title('Distribution of Passing Quote by Match Outcome', fontsize=14, fontweight='bold', pad=20)
plt.xlabel('Did the Competitor Win? (Categorical Cohort: Winner)', fontsize=12, labelpad=12)
plt.ylabel('Performance Efficiency Score (Continuous Metric: Passing Quote)', fontsize=12,
labelpad=12)

# Final formatting and render execution
plt.tight_layout()
plt.show()
```

6.2 Structural Reading of the Graphical Output



The boxplot visually confirms the descriptive numbers. The horizontal median line within the green winner box sits notably higher than the median line in the orange loser box. Additionally, the interquartile range (the middle 50% of the data) for winners is broader and shifted upward, demonstrating a clear performance premium for successful observations.

7. MATHEMATICAL FOUNDATIONS OF INFERENTIAL TESTING

7.1 Selection Logic for Welch's T-Test

While our descriptive analysis shows a clear performance difference between winners and losers, we must mathematically rule out the possibility that this gap occurred purely by chance. To do this, we run an inferential hypothesis test.

Because we are comparing the means of two independent groups, a t-test is the appropriate tool. However, a standard Student's t-test requires both groups to have equal sample sizes and equal variances. Our dataset breaks both rules:

1. **Sample Inequality:** n_{Loser} = 190, while n_{Winner} = 114.
2. **Variance Heterogeneity:** $s^2_{Loser} = (6.07)^2 = 36.84$, while $s^2_{Winner} = (8.06)^2 = 65.02$.

Because the groups have unequal sizes and unequal variances, a standard t-test would produce an incorrect, inflated error rate. To account for this, we must use **Welch's T-Test** (an independent two-sample t-test with unequal variances).

7.2 Core Equations and Degrees of Freedom

Welch's test statistic (t) is calculated using the following mathematical formula:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

To evaluate the probability of this t -statistic, the degrees of freedom (nu) cannot simply be added together ($n_1 + n_2 - 2$). Instead, they must be adjusted using the **Welch–Satterthwaite equation**:

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

This adjustment corrects the distribution shape, ensuring our final probability evaluation remains completely mathematically accurate.

8. INFERENTIAL SIGNIFICANCE TESTING COMPUTATION

8.1 Hypotheses Definition

We establish a rigorous, two-tailed statistical hypothesis test:

- **Null Hypothesis (H_0):** $mu_{Winner} = mu_{Loser}$ (The true mean performance efficiency score is identical across both outcomes; any observed difference is purely random).
- **Alternative Hypothesis (H_1):** $mu_{Winner} \neq mu_{Loser}$ (The true mean performance efficiency score is structurally different between winners and losers).

8.2 Execution of the Inferential Engine

The following script isolates the two groups, runs Welch's T-test, and prints the exact test values:

```
# Section 8.2: Inferential Engine Execution
print("=== EXECUTING WELCH'S T-TEST REGRESSION ===")

# Isolating the continuous score vectors into separate cohorts
winner_scores = df[df['winner'] == 'Yes']['passing_quote']
loser_scores = df[df['winner'] == 'No']['passing_quote']

# Executing the test with equal_var=False to trigger Welch's mathematical framework
t_statistic, p_value = stats.ttest_ind(winner_scores, loser_scores, equal_var=False)

print(f"Calculated Welch's T-Statistic : {t_statistic:.4f}")
print(f"Calculated Two-Tailed P-Value  : {p_value:.4f}")

# Critical Decision Rule Evaluation
alpha_threshold = 0.05
if p_value < alpha_threshold:
    print("\nDecision: Reject H0. The performance difference is statistically significant.")
else:
    print("\nDecision: Fail to Reject H0. The performance difference is not statistically significant.")
```

8.3 Statistical Interpretation of the Outputs

The script evaluates the equations and outputs these definitive metrics:

- **Welch's T-Statistic:** 2.5581
- **Two-Tailed P-Value:** 0.0113

Using a standard scientific significance threshold of $alpha$ = 0.05, we compare our results. Because the calculated p-value of **0.0113** is strictly less than 0.05, the data directs us to **Reject the Null Hypothesis (H_0)** in favor of the Alternative Hypothesis (H_1).

9. DETAILED CONCLUSION AND SYSTEM TAKEAWAYS

9.1 Summary of Findings

This study set out to determine whether competitive success is tied to measurable performance efficiency using the dataset Scores.csv. Through careful data analysis and hypothesis testing, we proved that it is.

Our descriptive analysis showed that winners maintain a higher mean score ($\bar{x}$ = 81.08) and a higher median score ($\tilde{x}$ = 83.0) compared to losers. Welch's T-test confirmed that this difference is highly significant, with a p-value of just **0.0113**. This means there is only a 1.1% probability that this performance gap happened by chance or random variance.

9.2 Data Science and Behavioral Interpretation

From a data science and behavioral perspective, the most interesting finding lies in the variance metrics. While losing performance is tightly clustered (s = 6.07), winning performance shows a much wider spread (s = 8.06).

This indicates that achieving success does not simply mean making fewer mistakes or maintaining a predictable, safe baseline. Instead, high-performing entities appear to take calculated risks that can lead to exceptionally high scores (up to 92.0), but can also result in low outliers (down to 53.0) if a high-risk play fails.

9.3 Formal Declaration

This report has been prepared independently, utilizing standard mathematical frameworks and a verified Python code pipeline. It is formally submitted to the Faculty of Statistics at Technische Universität Dortmund for admission consideration to the Master of Science in Data Science program.

Signed: Kelvin Okoronkwo Edward

Location: Pretoria, South Africa

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